

Background

Forest-based centrality on undirected graphs. Centrality measures based on forests have been proposed and extensively studied, which capture the influence of nodes and edges through their participation in spanning forests. In this paper, we focus on two forest-based centrality measures:

- Forest Node Centrality (FNC): The average size of the connected component containing node u over all spanning forests rooted at u , expressed as:

$$\text{FNC}(u) = \frac{\sum_{\phi \in \mathcal{F}_{uu}} |T_\phi(u)|}{|\mathcal{F}_{uu}|} = \frac{1}{\omega_{uu}}.$$

- Forest Edge Centrality (FEC): The average size of the connected component containing edge (u, v) over spanning forests that v is rooted at u , expressed as:

$$\text{FEC}(u, v) = \frac{1}{|\mathcal{F}_{vu}|} \sum_{\phi \in \mathcal{F}_{vu}, (u, v) \in \phi} |T_\phi(u)| = \frac{\omega_{uu} + \omega_{vv} - 2\omega_{uv}}{\omega_{uv}}.$$

Table 1. Notations.

Notation	Description
$\mathcal{G}$	An undirected graph $\mathcal{G}$
D	Degree diagonal matrix
A	Adjacency matrix
$L = D - A$	Laplacian matrix
$\Omega = (I + L)^{-1}$	Forest matrix with each entry ω_{uv}
$\mathcal{F}$	Set of all spanning forests
$\mathcal{F}_{uv}$	Spanning forests that u is rooted at v
$T_\phi(u)$	Set of nodes in the connected component containing u

Motivation

- Exact computation of FNC and FEC involves enumerating satisfying spanning forests or inefficient matrix inversion, which is impractical for large graphs.
- Existing methods for calculating forest matrix focus on directed graphs. The matrix on undirected graphs possess crucial properties, indicating that there is still room for improvement in algorithms applied to directed graphs.

Contribution

- We analyze the properties of spanning forests on **undirected graphs** and devise new estimators for the entries of the forest matrix.
- We combine the strengths of our designed estimator with the neighbor-based estimator in a **simple yet non-trivial way**, leading to a more accurate and faster estimator. Building on this, we develop the algorithm IFGN.
- Based on the definition of FEC, we identify a **new physical interpretation** that allows for better computation of FEC. Building on this approach and our **variance reduction technique** on undirected graphs, we introduce the algorithm FECE.

Fast Approximation Algorithm for FNC

Probabilistic interpretation

The entry ω_{uv} in the forest matrix represents the probability of the root of node u being node v in a uniformly sampled spanning forest.

Isomorphic Forest Group

Two forests ϕ_1 and ϕ_2 are considered isomorphic, denoted $\phi_1 \sim \phi_2$, if they have the same set of nodes and edges, differing only in their root nodes.

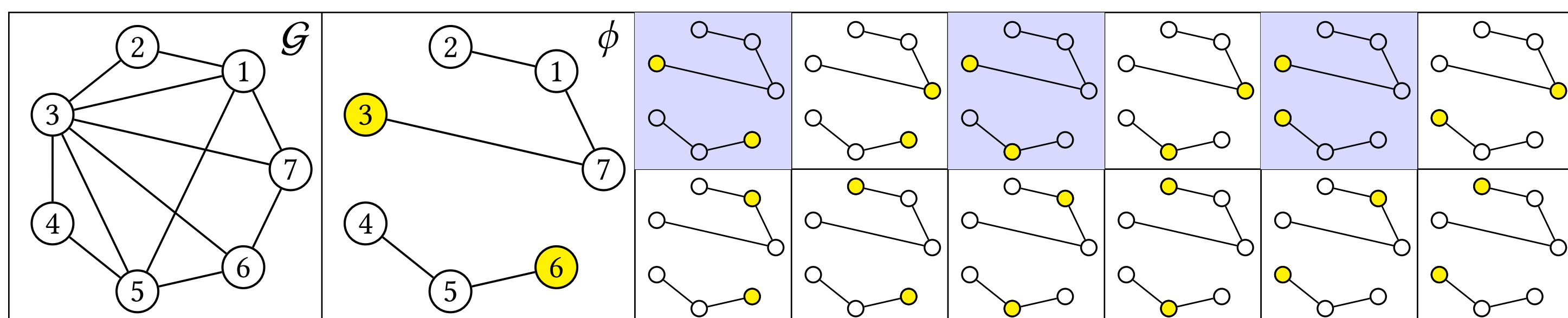


Figure 1. A toy graph $\mathcal{G}$, a spanning forest ϕ and its isomorphic forests.

Since Wilson's algorithm generates a uniformly spanning forest, if we sample a forest in which u and v appear in the same tree, we can consider that this forest contributes $\frac{1}{|T_\phi(u)|}$ to ω_{uv} for each node v in the connected component. Based on this, we propose a new estimator $\hat{\omega}_{uv}(\phi)$ for ω_{uv} , expressed as

$$\hat{\omega}_{uv}(\phi) = \mathbb{E}\{\hat{\omega}_{uv}(\phi_0) | \phi_0 \sim \phi\} = \frac{\mathbb{I}_{\{r_\phi(u)=r_\phi(v)\}}}{|T_\phi(u)|}.$$

Algorithm IFGN

- Employs the variance-reduced estimator of ω_{uv} based on **isomorphic forest group** on undirected graphs.
- Combines the variance reduction technique of **neighbor information** to provide a more precise estimate.
- Provides an error guarantee.

Time Complexity

- Algorithm IFGN have a time complexity of $O(nl\bar{d})$, where l is the number of sampled spanning forests.
- We find that its high accuracy allows for earlier termination of sampling with relatively fewer iterations due to the utilizing of Bernstein inequality.

Novel interpretation and Approximation for FEC

Forest Mapping Set

Every forest ϕ that has u as the root and includes the edge (u, v) can be mapped to $|T_\phi(u)|$ **isomorphic forests** that do not contain the edge (u, v) . Moreover, these forests precisely constitute the union of the sets $\mathcal{F}_{uu} \setminus \mathcal{F}_{vu}$ and $\mathcal{F}_{vv} \setminus \mathcal{F}_{uv}$.

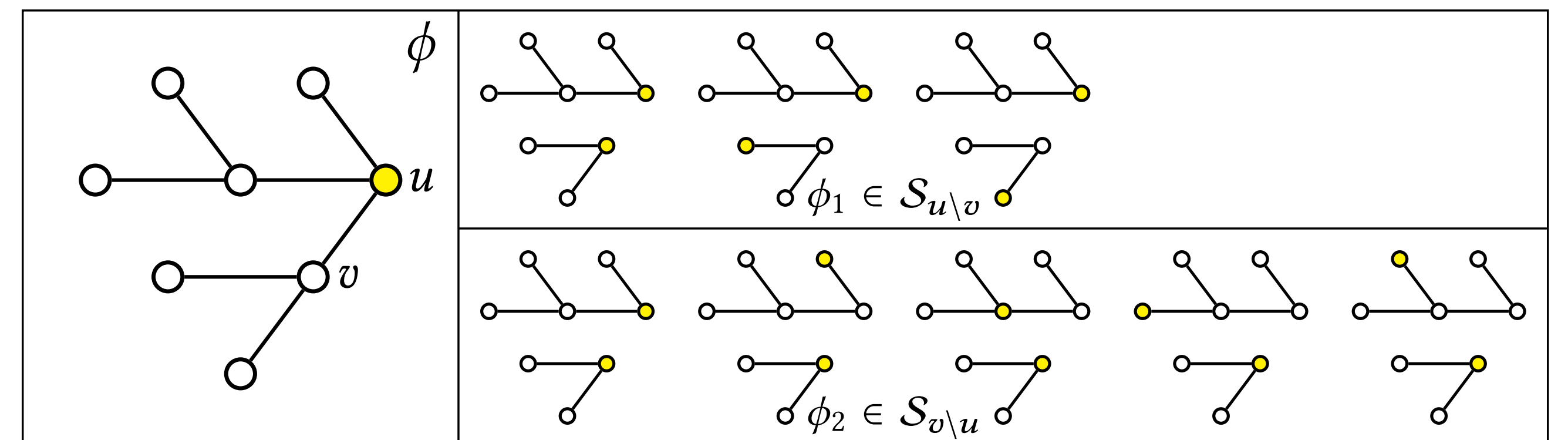


Figure 2. A spanning forest ϕ and its mapped set.

Algorithm FECE

- Incorporates a new physical interpretation of FEC based on **forest mapping set**, allowing for a better overall estimation.
- Runs in $O(nl)$ time and provides an ϵ -approximation for FEC.

Experiment

Accuracy Evaluation

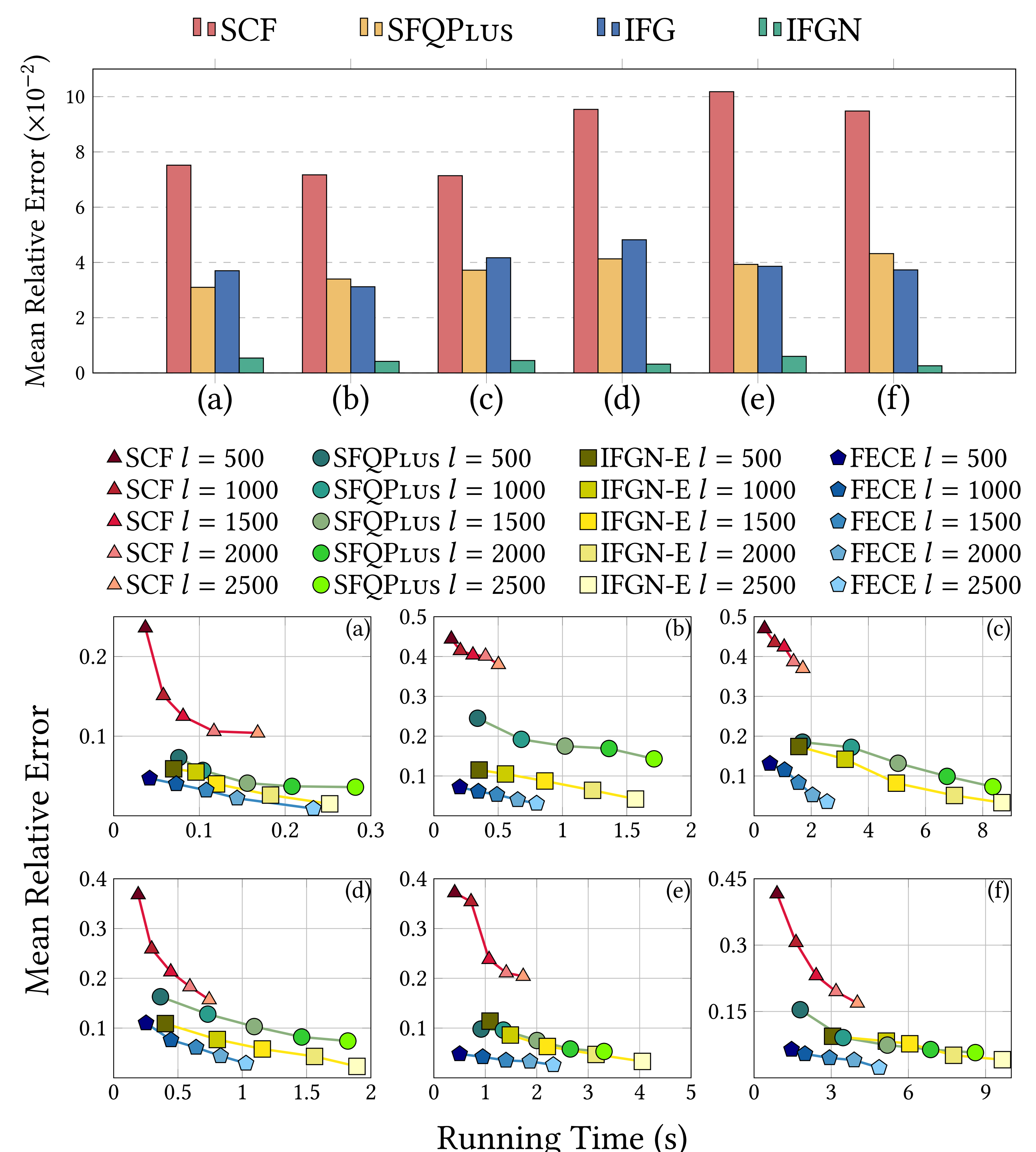


Figure 3. Mean relative error and running time for each approximation algorithm for FNC and FEC on six graphs: Bio-CE-LC (a), Hamsterster (b), Facebook (c), GrQc (d), WebBase (e) and Gnutella (f).

Efficiency Evaluation

Table 2. The running time (seconds) of approximating FNC and FEC on four large datasets: Youtube (a), US Patents (b), Dbpl (c) and CentralUSA (d).

Network	FNC				FEC		
	SCF	SFQPlus	IFG	IFGN	SCF	SFQPlus	FECE
Youtube	159.7	161.5	142.2	104.3	205.4	191.3	178.2
US Patents	989.9	1046	835.5	770.5	1695	1574	1385
Dbpl	1066	994.9	901.5	951.4	1342	1258	1041
CentralUSA	1982	2179	1526	1359	2476	2381	2205